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Econometrics 322

8 May 2026

## Determinants of Infidelity

## **Introduction**

Analyzing the individual factors that influence infidelity within marriages is a key topic in social science and economics. According to Pew Research Center, in 2023 about  $\frac{1}{3}$  of ever-married citizens' first marriage ended in divorce (Hayes). This statistic displays how divorce is at an all time high, and as a society we should take a closer look at factors that could impact this type of behavior, such as infidelity. Although divorce can be driven by different factors, adultery is a cause of divorce that has been around for decades. Understanding why people chose to be unfaithful and what characteristics lead to that behavior, can help explain patterns. This research is relevant to individuals, families, and policymakers, as understanding what leads to infidelity can help inform efforts to improve relationship stability and reduce divorce rates.

This empirical project aims to answer the following research question, “What factors influence the likelihood of having an affair, with a focus on marital satisfaction?”. The affairs dataset provided by the course page will help answer the main question. The ‘ratemarr’ which represents marital satisfaction on a scale of 1-5, is my main variable of interest. The control variables will be age, years married, presence of children, religiosity, and education. A logit model is used because the dependent variable, affair, is binary, taking a value 0 if the individual didn’t have an affair, and 1 otherwise. It is predicted that a lower marital satisfaction ranging from 1-3 will lead to a higher likelihood of infidelity, holding other variables constant.

## **Data**

The dataset used in this project is the affairs dataset which has 601 observations. The affairs dataset displays an individual's relationship behavior, marital satisfaction and demographic variables and how those affect the possibility of infidelity. As mentioned in the previous section, the dependent binary variable is affair (0=no affair, 1=affair). Since this study is

trying to estimate whether a person committed adultery or not, this is the outcome variable. Additionally, the independent variables are age, years married, kids, religiosity, education, and most importantly marital satisfaction. Although all these variables are diverse they can all help determine marriage stability and the likelihood of unfaithfulness. Age can dictate how mature a person may be, whereas years married can hint at marriage security. Couples who have more children are less likely to be disloyal, as it can disrupt multiple lives. Next, religiosity can judge if a couple has the same values/morals and education can suggest an individual's social behavior. Lastly, marital satisfaction clearly highlights how an individual feels in their relationship. The lower the overall satisfaction, the more likely an individual is to be unfaithful.

## Methodology

To explain the relationship between the dependent and independent variables and predict the probability of infidelity, we will be using a logit model. Using a logit model is most appropriate because the dependent variable, affair, is binary.

$$\text{logit}(P(\text{affair})) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{yrsmarr} + \beta_3 \text{kids} + \beta_4 \text{relig} + \beta_5 \text{educ} + \beta_6 \text{ratemar}_2 + \beta_7 \text{ratemarr}_3 + \beta_8 \text{ratemarr}_4 + \beta_9 \text{ratemarr}_5$$

Rather than treating marital satisfaction (ratemarr) as a continuous variable, it is modeled using a set of dummy variables representing each satisfaction category. Allowing for a more flexible specification since the categories may not follow a linear scale. The coefficients represent each factor's influence on the log-odds of having an affair. A positive coefficient would symbolize the possibility of adultery, and a negative coefficient would represent otherwise. To start off variables, age and years married may vary and are ambiguous because relationships can experience increased steadiness, or possible dissatisfaction. Additionally, education is also ambiguous due to differentiating opportunities and social interactions. Moreover, with the

presence of one or more kids we predict there is an expected negative sign, as disloyalty and children have an inverse relationship. Lastly, religion and marital satisfaction also have an expected negative sign due to the negative correlation with infidelity. The higher marital happiness, and similar religiosity levels a relationship has, the lower chance of unfaithfulness. After regressing the model we will analyze based on coefficient sign and statistical significance. Although the model gives insight into controlled variables affecting infidelity, it is important to talk about omitted variables bias. As stated above, an example of a threat is an individual's income, which is not included in the regression model, but can influence marital satisfaction and infidelity. Due to the restrictions of the dataset and logit model, techniques such as panel data and instrumental variables are not appropriate for this analysis.

## Results:

Looking at the results of the model, a logit model was used to estimate and look at the factors influencing infidelity occurring while there being a focus on marital satisfaction.

$$\begin{aligned} \text{logit}(P(\text{affair})) = & 0.2744(0.9174) - 0.0387(0.0178)\text{age} + 0.0893(0.0318)\text{yrsmarr} \\ & + 0.3617(0.2926)\text{kids} - 0.3210(0.0899)\text{relig} + 0.0532(0.0427)\text{educ} \\ & - 0.0079(0.5810)\text{ratemarr2} - 0.8101(0.5749)\text{ratemarr3} - 1.1189(0.5564)\text{ratemarr4} \\ & - 1.6065(0.5656)\text{ratemarr5} \end{aligned}$$

The result suggests that marital satisfaction (ratemarr) is a key factor of infidelity. The coefficients on the marital satisfaction categories are generally negative relative to the reference group, specifically 'ratemarr4' and 'ratemarr5' indicating that higher levels of marital satisfaction are associated with a lower probability of engaging in infidelity. While holding the other variables constant. This result or finding shows that the initial hypothesis that individuals who report lower satisfaction in their marriage are more likely to be unfaithful.

However, looking at the controlled variables, the effects show that they are mixed, meaning that age has a negative relationship with infidelity, whereas years married has a positive correlation. This means older people are less likely to be unfaithful, however longer marriages lead to higher chances of infidelity. The variable kids and educ is associated with an inconclusive stance because there is not enough evidence to conclude that children and a standard of education leads to infidelity. Looking at the variable 'relig', it has a negative relationship with infidelity because this shows that people who have stronger religious values are less likely to be having an affair. Overall, the results highlight that marital satisfaction is one of the most important predictors in infidelity in this specific model.

**Table 1**

| <b>Variable</b> | <b>Marginal Estimate (AME)</b> |
|-----------------|--------------------------------|
| age             | -0.0065                        |
| educ            | 0.0089                         |
| kids            | 0.0603                         |
| ratemarr2       | -0.0019                        |
| ratemarr3       | -0.1775                        |
| ratemarr4       | -0.2342                        |
| ratemarr5       | -0.3078                        |
| relig           | -0.0536                        |
| yrs marr        | 0.0149                         |

Table 1 highlights the marginal estimates from the logit model. The marginal estimate of a logit model shows the estimated change in the likelihood of infidelity correlated with a one-unit change in each variable while maintaining the same values for the other variables. The values

coincide with the standard error and coefficient values as shown in the regression equation above. The findings from it show that dissatisfaction within marriage increases the likelihood of finding fulfillment outside of the relationship, which provides strong support to the research question.

### **Discussion:**

This research question and model were set to examine the factors that influence the possibility of infidelity, with a specific focus on marital stratification. Using and applying the logit model to the affairs dataset, the analysis focused on having the understanding how both individual characteristics and relationship dynamics contribute to the probability of having an affair. These findings showed that while the multiple factors play a role, marital satisfaction comes out to being the most important predictors for infidelity while supporting the question. The results align with both economic and social theories of behavior because the relationships most likely are to be stayed committed with the current relationship all to show that it provides a greater overall satisfaction or utility. However, if the marital satisfaction is low, then there would have to be a finding for alternatives that increases, meaning that it could manifest as infidelity. This helps explain why marital satisfaction is the key variable of the model and the idea that emotional fulfillment and relationship quality are important components of marital stability. With all the variables that are included these allow for a more better understanding of the factors that surround infidelity. Some variables such as religiosity act more as stabilizing forces but ages and years married show more of a less predictable effect as well as being complex. There are many more insights that these variables prove, but it helps us understand where the nature of relationships have both personal characteristics and external responsibilities that shape behavior in different ways. The inclusion of control variables provides a better understanding of the

factors surrounding infidelity. Despite these insights, there is a limitation that could be considered, which would be omitted variable bias, but this would not work for the logit model because it would become too messy and make it more difficult to look at the results. Also, the use of cross sectional data limits the ability to draw conclusions. While the analysis identifies a strong association between marital satisfaction and infidelity, it cannot fully determine the direction of the relationship. For example, infidelity may contribute to lower reported satisfaction rather than reverse. Overall, this study provides insight into the role of marital satisfaction in influencing infidelity while acknowledging the complexity of relationships. Future research could expand on this analysis by incorporating additional variables or using the longitudinal data to better capture how relationships evolve over time.

**Conclusion:**

In conclusion, this study examined the factors influencing the likelihood of infidelity, with a primary focus on marital satisfaction. Using the logit model and the affairs dataset, the results show that marital stratification is a key predictor of unfaithful behavior, with lower satisfaction associated with a higher probability of having an affair. While other variables such as religiosity also contribute to the possibility of infidelity, their effects are less consistent than compared to the strong influence of relationship quality. Overall, the analysis suggests improving marital satisfaction may play a crucial role in reducing the chance of unfaithful behavior.

## Works Cited

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